

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : S, I(A) & LIVE-I(A))
ANSWER KEY
PAPER-1
PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C	A,C,D	B,C	B,D	B,D	A,C	B,C	A,D
SECTION-I (ii)	Q.	9	10	11	12				
	A.	A	B	D	B				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	18.00	2.66 to 2.67	48.00	20.00	3.50	9.00		

PART-2 : CHEMISTRY

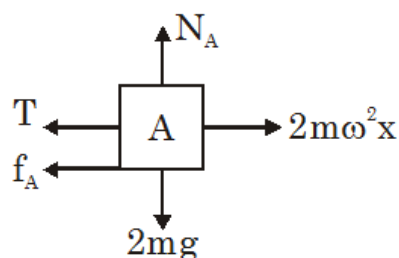
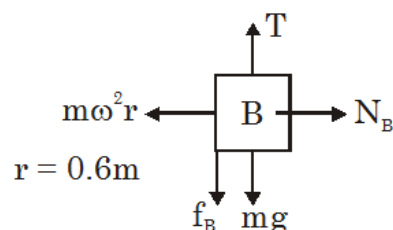
SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	B,C,D	A,B,D	A,C	B,C	B,D	B,D	A,B,D	A,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	C	D	C				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	4.00	5.00	3.00	9.60	240.00	2.00		

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6	7	8
	A.	A,C,D	A,B,C,D	D	A,B	A,B,C,D	A	A,B,C,D	A,B,C
SECTION-I (ii)	Q.	9	10	11	12				
	A.	D	B	B	A				
SECTION-II	Q.	1	2	3	4	5	6		
	A.	40.00	0.50	4.00	7.00	4.00	3.00		

HINT – SHEET
PART-1 : PHYSICS
SECTION-I (i)

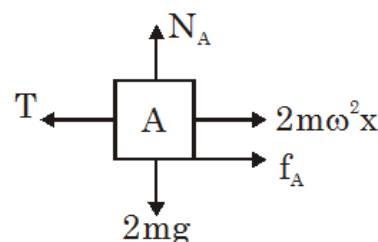
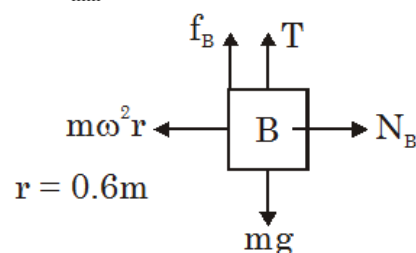
1. Ans (A,C)

 For $x_{\max} \Rightarrow$


$$2m\omega^2 x = T + f_A \quad \dots (i)$$

$$T = f_B + mg \quad \dots (ii)$$

$$f_A = \mu N_A, f_B = \mu N_B$$

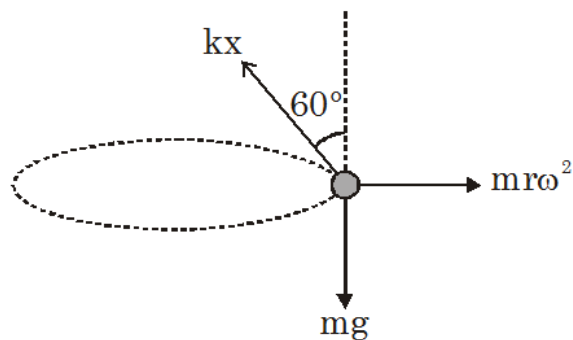
 For $x_{\min} \Rightarrow$


$$f_A + 2m\omega^2 x = T \quad \dots (i)$$

$$T + f_B = mg \quad \dots (ii)$$

$$f_A = \mu N_A, f_B = \mu N_B$$

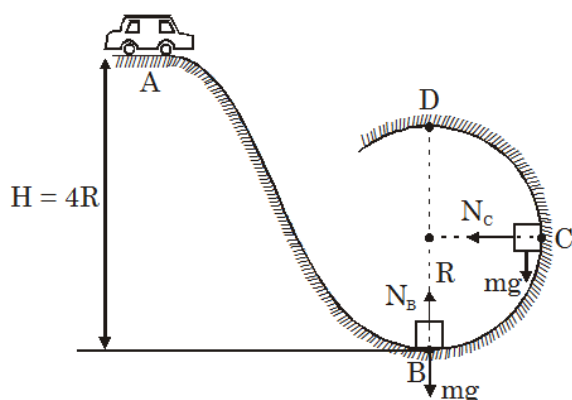
2. Ans (A,C,D)



$$kx \cdot \cos 60 = mg$$

$$kx \sin 60 = mr\omega^2$$

3. Ans (B,C)



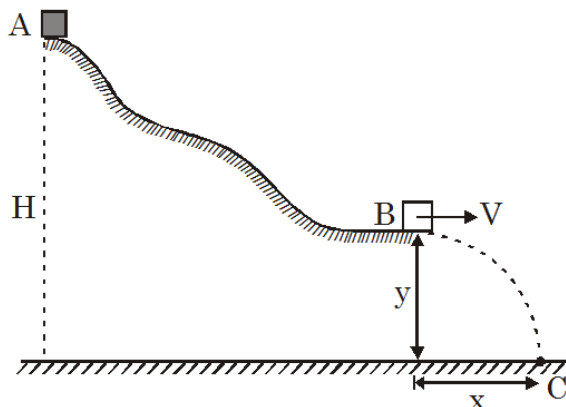
$$V_B = \sqrt{2gH} = \sqrt{8gR}$$

$$V_C = \sqrt{2g(3R)} = \sqrt{6gR}$$

$$N_B - mg = \frac{mV_B^2}{R}$$

$$N_C = \frac{mV_C^2}{R}$$

4. Ans (B,D)



$$mg(H - y) = \frac{1}{2}mv^2$$

$$v = \sqrt{2g(H - y)}$$

$$t = \sqrt{\frac{2y}{g}}$$

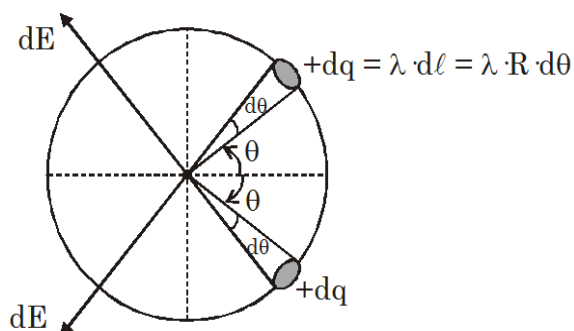
$$x = vt = 2\sqrt{y \cdot (H - y)}$$

$$\text{For } x_{\max} \Rightarrow \frac{dx}{dy} = 0$$

$$y = \frac{H}{2}$$

$$x_{\max} = H$$

5. Ans (B,D)

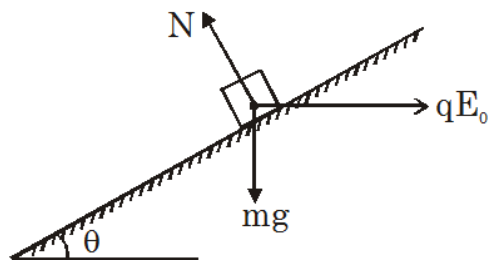


$$E = \int_{-\pi/2}^{\pi/2} 2 \cdot dE \cos \theta$$

$$= 2 \int \frac{K \cdot dq}{R^2} \cdot \cos \theta$$

$$= 2 \int_{-\pi/2}^{\pi/2} \frac{K \cdot \lambda_0 \cos \theta \cdot R \cdot d\theta \cdot \cos \theta}{R^2}$$

6. Ans (A,C)



$$N = mg \cos \theta + qE_0 \sin \theta$$

$$mg \sin \theta = qE_0 \cos \theta$$

7. Ans (B,C)

$$\frac{1}{10} = (1.5 - 1) \left(\frac{2}{R} \right)$$

$$R = 10 \text{ cm}$$

$$P = \frac{1.5 - 1}{10} + \frac{\frac{4}{3} - 1.5}{-10}$$

$$\frac{4}{3v} - \frac{1}{-21} = \frac{1}{20} + \frac{1}{60}$$

8. Ans (A,D)

$$\sin i_g = \mu \sin (A - \theta_C)$$

$$= \mu [\sin A \cos \theta_C - \cos A \sin \theta_C]$$

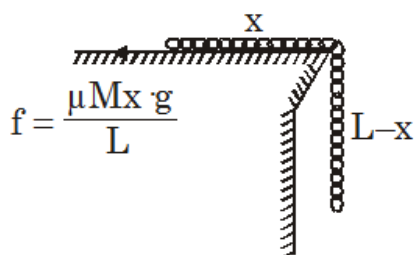
$$\sin i_g = \mu \left[\sin A \cdot \sqrt{1 - \frac{1}{\mu^2}} - \cos A \cdot \frac{1}{\mu} \right]$$

$$i_g = 30$$

PART-1 : PHYSICS

SECTION-II

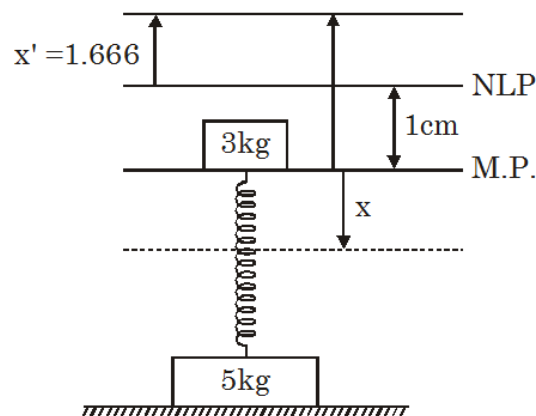
1. Ans (18.00)



$$f = \frac{\mu M x g}{L}$$

$$\omega = - \int_{\frac{L}{3}}^0 \frac{\mu M}{L} g x \cdot dx = \frac{-\mu M g L}{18}$$

2. Ans (2.66 to 2.67)



$$k \times \frac{1}{100} = 30$$

$$k = 3000$$

$$kx' = 50$$

$$x' = \frac{50}{3000}$$

$$= \frac{5}{3}$$

$$= 1.666 \text{ cm}$$

$$x = 2.666$$

3. Ans (48.00)

$$P = \left(\frac{\mu_A - 1}{\alpha} \right) \cdot 2 + \left(\frac{\mu_B - \mu_A}{-R} \right) \cdot 2 + \left(\frac{-2\mu_B}{-\frac{R}{2}} \right)$$

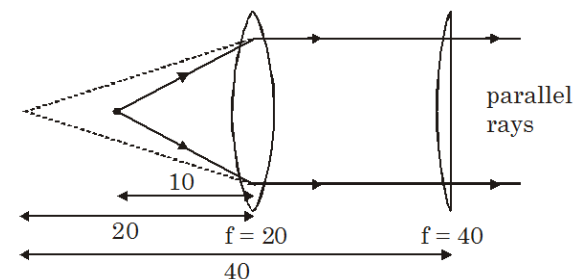
And

$$P = -\frac{1}{f}$$

$$\text{From } \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$$

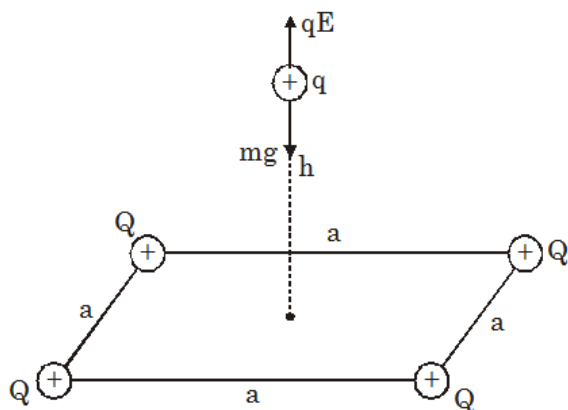
$$\frac{1}{-40} + \frac{1}{-10} = \frac{1}{f}$$

4. Ans (20.00)



$$d = 20 \text{ cm}$$

5. Ans (3.50)



$$qE = mg$$

$$\frac{q \cdot KQ \cdot h}{\left[h^2 + \left(\frac{a}{\sqrt{2}} \right)^2 \right]^{\frac{3}{2}}} = mg$$

6. Ans (9.00)

$$x = 3 \sin(100t) + 8 \cos^2(50t)$$

$$x = 3 \sin(100t) + 8 \left[\frac{\cos(100t) + 1}{2} \right]$$

$$= 4 + 3 \sin(100t) + 4 \cos(100t)$$

$$= 4 + 5 \sin(100t + 53^\circ)$$

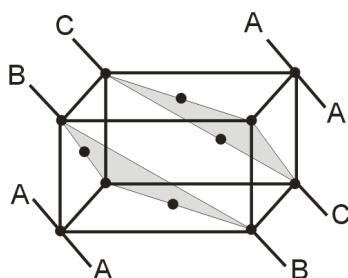
$$x_{\max} = 9 \text{ cm}$$

PART-2 : CHEMISTRY

SECTION-I (i)

4. Ans (B,C)

- (A) SC involve 2-D square arrangement of atoms.
(B) Correct,



- (C) Correct, HCP can also be formed by ABABABAB arrangements where A and B 2d hexagonal arrangements.
(D) False.

5. Ans (B,D)

$$C = \frac{600}{7} \% \Rightarrow \frac{600}{12 \times 7} = \frac{50}{7}$$

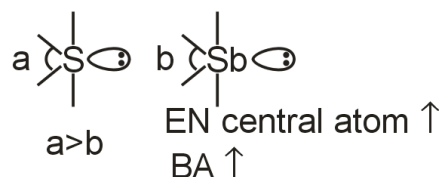
$$H = \frac{100}{7} \% \Rightarrow \frac{100}{1 \times 7} = \frac{50}{7}$$

$$\therefore \frac{C}{H} \text{ ratio} = \frac{1}{2} \Rightarrow \text{EF} = \text{CH}_2$$

$$\text{Molar mass} = 93.33 \times 10^{-24} \times 6 \times 10^{23}$$

$$\therefore \text{molecule formula} = \text{C}_4\text{H}_8$$

6. Ans (B,D)



PART-2 : CHEMISTRY

SECTION-I (ii)

9. Ans (D)

$$\text{By definition, } \alpha = \frac{C_{A_0} - C_A}{C_{A_0}}$$

$$\text{For zero order, } C_A = C_{A_0} - kt \Rightarrow \alpha = \frac{kt}{C_{A_0}}$$

$$\text{For 1st order, } C_A = C_{A_0} e^{-kt}$$

$$\Rightarrow C_{A_0}(1 - \alpha) = C_{A_0} e^{-kt}$$

$$\Rightarrow \alpha = 1 - e^{-kt}$$

$$\text{For 2nd order, } \frac{1}{C_A} = \frac{1}{C_{A_0}} + kt$$

$$\text{or, } \frac{C_{A_0} - C_A}{C_A C_{A_0}} = kt$$

$$\text{or, } \alpha = \frac{C_{A_0} kt}{1 + C_{A_0} kt}$$

PART-2 : CHEMISTRY

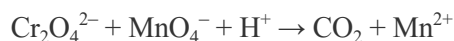
SECTION-II

4. Ans (9.60)

$$\pi = CRT = 0.4 \text{ M} \times 0.08 \frac{\text{L} \cdot \text{atm}}{\text{molK}} \times 300 \text{ K} = 9.6 \text{ atm}$$

5. **Ans (240.00)**

0.5 M KMnO_4



v.f. = 2 v.f. = 5, 0.5 M

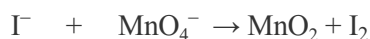
1.5 M, V_1 ml

125 M

eq. of MnO_4^- = eq. of $\text{C}_2\text{O}_4^{2-}$

or, $0.5 \times V_1 \times 5 = 1.5 \times 125 \times 2$

or, $V_1 = 150$ ml



0.5 M, 0.5M

270 ml

v.f.=1 v.f.=3

V_2 ml

eq. of I^- = eq. of MnO_4^-

or, $0.5 \times 270 \times 1 = 0.5 \times V_2 \times 3$

or, $V_2 = 270$ ml

$\therefore$ Total volume of $\text{KMnO}_4 = V_1 + V_2 = 150 + 90 = 240$ ml

6. **Ans (2.00)**

s-p orbit mixing will be in 14 or urs proton

system

O_2^- para

N_2 dia

C_2 dia

CN^- dia

B_2 para

O_2^{+2} dia

PART-3 : MATHEMATICS

SECTION-I (i)

3. **Ans (D)**

$$f(x) = \begin{cases} (x+1)2^{-\frac{2}{x}}, & x > 0 \\ 0, & x = 0 \\ x+1, & x < 0 \end{cases}$$

$$\text{RHL} = \lim_{x \rightarrow 0^+} (x+1)2^{-\frac{2}{x}} = 0$$

$$\text{LHL} = \lim_{x \rightarrow 0^-} (x+1) = 1$$

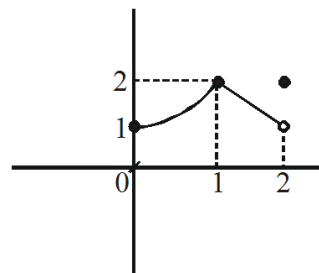
$$\Rightarrow \lim_{x \rightarrow 0^+} f(x) \neq \lim_{x \rightarrow 0^-} f(x)$$

4. **Ans (A,B)**

$$f(x) = x^2 + 1 \quad 0 \leq x < 1$$

$$f(x) = 3 - x \quad 1 \leq x < 2$$

$$f(x) = 2 \quad x = 2$$



option A & B are correct.

5. **Ans (A,B,C,D)**

Case-1

Let $f(x)$ is discontinuous

$$f(1^-) \neq f(1^+)$$

$$-2 \neq a + 1$$

$$a \neq -3$$

and $g(x)$ is continuous at $x = 2$

$$f(2^-) = f(2^+)$$

$$0 = 1 - b$$

$$b = 1$$

Case-2

Let $f(x)$ is continuous at $x = 1$

$$f(1^-) = f(1^+)$$

$$-2 = a + 1$$

$$a = -3$$

and $g(x)$ is discontinuous at $x = 2$

$$f(2^-) \neq f(2^+)$$

$$0 \neq 1 - b$$

$$b \neq 1$$

6. **Ans (A)**

$$g'(x+y) = e^y g'(x) + e^x g(y)$$

$$g'(y) = e^y g'(0) + g(y)$$

$$g'(y) = 2e^y + g(y) \quad \dots (1)$$

$$\therefore g'(5) - 2e^5 = g(5)$$

$$\text{from (1)} \quad g'(y) - g(y) = 2e^y$$

$$e^{-y} g'(y) - e^{-y} g(y) = 2$$

$$\frac{d}{dy} (e^{-y} g(y)) = 2$$

$$e^{-y} g(y) = 2y + c$$

$$g(x) = (2x + c)e^x$$

$$\text{since } g(0) = 0 \Rightarrow g(x) = 2xe^x$$

$$\therefore g(5) = 10e^5$$

7. **Ans (A,B,C,D)**

$$\sin^{-1}x \in \left[-\frac{\pi}{2}, \frac{\pi}{2}\right] \quad \forall x \in [-1, 1]$$

$$0 \leq (\sin^{-1}x)^2 \leq \frac{\pi^2}{4}$$

$$0 \leq (\sin^{-1}y)^2 \leq \frac{\pi^2}{4}$$

$$0 \leq (\sin^{-1}z)^2 \leq \frac{\pi^2}{4}$$

$$\sum (\sin^{-1}x)^2 = \frac{3\pi^2}{4} \quad \text{when } x = y = z = 1 \text{ or } -1$$

$(x - y + z)$ can be 3, -3, -1, 1.

8. **Ans (A,B,C)**

For any 'x' which is rational $f(x) = \pi$
so many one function.

Range of $f(x)$ is $\{-\pi, \pi\}$ so function is into.

Here $f(x) = f(-x) \quad \forall x \in \mathbb{R}$

so $f(x)$ is an even function

If x is rational then $f(f(x)) = f(\pi) = -\pi$

If x is irrational then $f(f(x)) = f(-\pi) = -\pi$

Hence $f(f(x)) = -\pi \quad \forall x \in \mathbb{R}$

PART-3 : MATHEMATICS

SECTION-I (ii)

9. **Ans (D)**

$$f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

$$f'(x) = 3x^2 + 2$$

$$f(x) = x^3 + 2x + c$$

$$f(0) = 1 \Rightarrow c = 1$$

$$\therefore f(x) = x^3 + 2x + 1$$

$$f'(1) = 5$$

$$g(x) = e^{-x} f(x)$$

$$g'(x) = e^{-x} f'(x) - e^{-x} f(x)$$

$$g'(3) = e^{-3} f'(3) - e^{-3} f(3) = -5e^{-3}$$

10. **Ans (B)**

$$f'(x) = \lim_{h \rightarrow 0} \left(\frac{f(x+h) - f(x)}{h} \right)$$

$$f'(x) = 3x^2 + 2$$

$$f(x) = x^3 + 2x + c$$

$$f(0) = 1 \Rightarrow c = 1$$

$$\therefore f(x) = x^3 + 2x + 1$$

$$f'(1) = 5$$

$$g(x) = e^{-x} f(x)$$

$$g'(x) = e^{-x} f'(x) - e^{-x} f(x)$$

$$g'(3) = e^{-3} f'(3) - e^{-3} f(3) = -5e^{-3}$$

11. **Ans (B)**

$$(I) \tan \left(\tan^{-1} \frac{1}{x} + \tan^{-1} \frac{1}{y} \right) = \tan \left(\tan^{-1} \frac{1}{10} \right)$$

$$\frac{\frac{1}{x} + \frac{1}{y}}{1 - \frac{1}{xy}} = \frac{1}{10} \Rightarrow (x-10)(y-10) = 101$$

$$(x, y) \equiv (11, 111), (111, 11), (9, -91), (-91, 9)$$

$\therefore$ 4 ordered pairs

$$(I) \rightarrow (R, S, T, U)$$

(II)

$$\tan \left(\tan^{-1} \left(\frac{1}{1+2.3} \right) + \tan^{-1} \left(\frac{1}{1+3.4} \right) + \dots + \tan^{-1} \left(\frac{1}{1+19.20} \right) \right)$$

$$\tan((\tan^{-1}3 - \tan^{-1}2) + (\tan^{-1}4 - \tan^{-1}3) + \dots +$$

$$(\tan^{-1}20 - \tan^{-1}19))$$

$$\tan(\tan^{-1}20 - \tan^{-1}2) = \tan \left(\tan^{-1} \left(\frac{18}{1+40} \right) \right)$$

$$= \frac{18}{41} = \frac{m}{n}$$

$$(m+n+1) = 60$$

$$(II) \rightarrow (P, Q, R, U)$$

$$(III) \lim_{x \rightarrow 0} \left(\frac{x}{\sin x} \right)^{\frac{1}{x^2}} \quad (1^\infty \text{ form})$$

$$e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{x}{\sin x} - 1 \right)}$$

$$e^{\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^2 \cdot \frac{\sin x}{x} \cdot x} \right)}$$

$$e^{\lim_{x \rightarrow 0} \frac{x - \left(x - \frac{x^3}{3!} \right)}{x^3 \cdot \left(\frac{\sin x}{x} \right)} = e^{\left(\frac{1}{6} \right)} = e^{\left(\frac{a}{b} \right)}$$

$$\therefore a = 1$$

$$(III) \rightarrow (P, Q, R, S, T, U)$$

$$(IV) \lim_{x \rightarrow 0} \left(\left[\frac{3 \tan^{-1} x}{x} \right] + \left[\frac{2 \sin x}{x} \right] + 2 \left[x \operatorname{cosec} x \right] \right)$$

$$= (2 + 1 + 2) = 5$$

$$(IV) \rightarrow (P, Q, R)$$

12. Ans (A)

$$(I) \tan\left(\tan^{-1}\frac{1}{x} + \tan^{-1}\frac{1}{y}\right) = \tan\left(\tan^{-1}\frac{1}{10}\right)$$

$$\frac{\frac{1}{x} + \frac{1}{y}}{1 - \frac{1}{xy}} = \frac{1}{10} \Rightarrow (x-10)(y-10) = 101$$

$$(x, y) \equiv (11, 111), (111, 11), (9, -91), (-91, 9)$$

$\therefore$ 4 ordered pairs

$$(I) \rightarrow (R, S, T, U)$$

(II)

$$\tan\left(\tan^{-1}\left(\frac{1}{1+2.3}\right) + \tan^{-1}\left(\frac{1}{1+3.4}\right) + \dots + \tan^{-1}\left(\frac{1}{1+19.20}\right)\right)$$

$$\tan((\tan^{-1}3 - \tan^{-1}2) + (\tan^{-1}4 - \tan^{-1}3) + \dots + (\tan^{-1}20 - \tan^{-1}19))$$

$$\tan(\tan^{-1}20 - \tan^{-1}2) = \tan\left(\tan^{-1}\left(\frac{18}{1+40}\right)\right)$$

$$= \frac{18}{41} = \frac{m}{n}$$

$$(m+n+1) = 60$$

$$(II) \rightarrow (P, Q, R, U)$$

$$(III) \lim_{x \rightarrow 0} \left(\frac{x}{\sin x}\right)^{\frac{1}{x^2}} \quad (1^\infty \text{ form})$$

$$e^{\lim_{x \rightarrow 0} \frac{1}{x^2} \left(\frac{x}{\sin x} - 1\right)}$$

$$e^{\lim_{x \rightarrow 0} \left(\frac{x - \sin x}{x^2 \cdot \frac{\sin x}{x} \cdot x}\right)}$$

$$e^{\lim_{x \rightarrow 0} \frac{x - \left(x - \frac{x^3}{3!}\right)}{x^3 \cdot \left(\frac{\sin x}{x}\right)}} = e^{\left(\frac{1}{6}\right)} = e^{\left(\frac{a}{b}\right)}$$

$$\therefore a = 1$$

$$(III) \rightarrow (P, Q, R, S, T, U)$$

$$(IV) \lim_{x \rightarrow 0} \left(\left[\frac{3\tan^{-1}x}{x}\right] + \left[\frac{2\sin x}{x}\right] + 2[x\operatorname{cosec}x]\right)$$

$$= (2 + 1 + 2) = 5$$

$$(IV) \rightarrow (P, Q, R)$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (40.00)

$$\ln(x+y) = 4xy \quad (\text{At } x=0, y=1)$$

$$x+y = e^{4xy}$$

$$\Rightarrow 1 + \frac{dy}{dx} = e^{4xy} \left(4x \frac{dy}{dx} + 4y\right)$$

$$\text{At } x=0 \quad \boxed{\frac{dy}{dx} = 3}$$

$$\frac{d^2y}{dx^2} = e^{4xy} \left(4x \frac{dy}{dx} + 4y\right)^2 + e^{4xy} \left(4x \frac{d^2y}{dx^2} + 4y\right)$$

$$\text{At } x=0, \frac{d^2y}{dx^2} = e^0(4)^2 + e^0(24)$$

$$\Rightarrow \frac{d^2y}{dx^2} = 40$$

2. Ans (0.50)

Differentiating w.r.t. x only

$$\frac{1}{6} f' \left(\frac{x+5y}{6} \right) = \frac{1}{6} f'(x)$$

$$\text{Put } x=0$$

$$f' \left(\frac{5y}{6} \right) = f'(0) = 1$$

$$\therefore f'(y) = 1$$

$$f(y) = y + c$$

$$\therefore f(y) = y - 1$$

$$f(2) = 1$$

4. Ans (7.00)

$$\text{At } x = \pm \frac{3}{2}, \pm 1, \pm \frac{1}{2}, 0$$

$$f(x) = \text{diff} + \text{non-diff} = \text{non-diff}$$

5. Ans (4.00)

$$\underbrace{(4\cos^2 x + \sec^2 x)}_{\geq 4} + \underbrace{(\tan^2 x + \cot^2 x)}_{\geq 2} = 6$$

$$\therefore \cos^2 x = \frac{1}{2} \text{ and } \tan^2 x = 1$$

Hence 4 solutions

6. Ans (3.00)

$$\tan^2 x + \tan^2(60^\circ - x) + \tan^2(60^\circ + x)$$

$$= 6 + 9\tan^2 3x$$

$$\text{So required sum is } 6 + 9 \tan^2 60 = 33$$

JEE(Main + Advanced) : ENTHUSIAST COURSE (PHASE : S, I(A) & LIVE-I(A))
ANSWER KEY
PAPER-2
PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	A	A	D	C	B	A	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	B	B	B	B	B	B	A	B	B	B
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	16	48	6	5	4	15	3	40	10	4

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	B	C	B	D	A	B	B	C	A	B
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	B	C	C	C	A	B	A	C	C
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	5	4	3	4	384	30	8	0	3	4

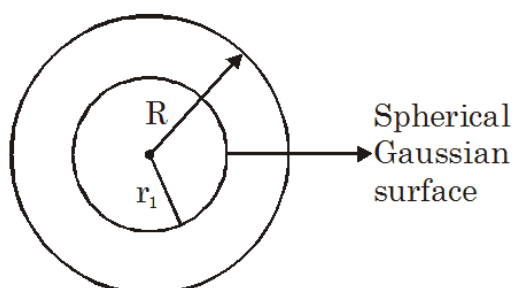
PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6	7	8	9	10
	A.	A	D	C	B	A	C	D	B	B	A
	Q.	11	12	13	14	15	16	17	18	19	20
SECTION-II	A.	A	D	D	C	B	C	C	C	B	A
	Q.	1	2	3	4	5	6	7	8	9	10
SECTION-II	A.	4	5	9	1	2	2	1	2	3	3

HINT – SHEET
PART-1 : PHYSICS
SECTION-I

1. Ans (A)

$$\oint \vec{E} \cdot d\vec{s} = \frac{q_{in}}{\epsilon_0}$$



$$E \cdot 4\pi r_1^2 = \frac{q_{in}}{\epsilon_0}$$

$$E = \frac{q_{in}}{4\pi\epsilon_0 r_1^2}$$

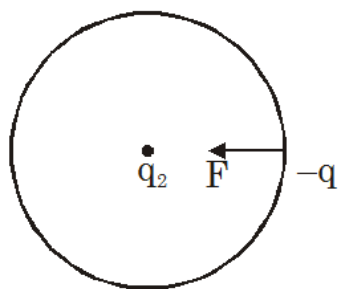
Where

$$q_{in} = \int_0^{r_1} \rho dV = \int_0^{r_1} \rho_0 \left(1 - \frac{r}{R}\right) 4\pi r^2 dr$$

$$= 4\pi\rho_0 \left(\frac{r_1^3}{3} - \frac{r_1^4}{4R}\right)$$

$$\therefore \text{From (i) } E = \frac{\rho_0 r_1}{3\epsilon_0} \left(1 - \frac{3r_1}{4R}\right)$$

2. Ans (A)



$$F = F_c$$

$$\frac{kq_1q_2}{r^2} = \frac{mv^2}{r}$$

$$v = \sqrt{\frac{q_1q_2}{4\pi\epsilon_0rm}}$$

$$T = \frac{2\pi r}{v} = 2\pi r \sqrt{\frac{4\pi\epsilon_0m}{q_1q_2}}$$

$$T = \sqrt{\frac{16\pi^3m\epsilon_0r^3}{q_1q_2}}$$

3. Ans (A)

$$\phi_1 = -E_0 (4a^2) \times a^2 = -E_0 4a^4$$

$$\phi_2 = E_0 (9a^2) \times a^2 = E_0 9a^4$$

$$\text{Total flux } \phi = 5E_0 a^4; \frac{q}{\epsilon_0} = 5E_0 a^4$$

$$q = 5\epsilon_0 a^4 E_0$$

4. Ans (D)

$$x = 10 \sin\left(4t - \frac{\pi}{3}\right)$$

$$\Rightarrow 6 = 10 \sin\left(4t - \frac{\pi}{3}\right)$$

$$\Rightarrow \left(4t - \frac{\pi}{3}\right) = 37^\circ$$

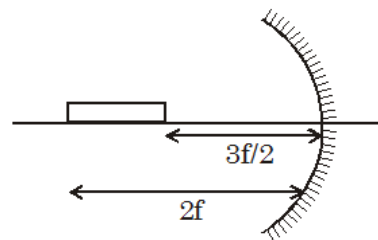
$$\therefore v = 10 \cos\left(4t - \frac{\pi}{3}\right) \times 4$$

$$= 10 \times \frac{4}{5} \times 4 = 32 \text{ m/s}$$

5. Ans (C)

Field inside spherical cavity is uniform.

6. Ans (B)



$$u = -\frac{3f}{2}$$

$$f = -f$$

$$\frac{1}{v_1} + \frac{1}{u} = \frac{1}{f}$$

$$v_1 = -3f$$

$$v_2 = -2f$$

$$\text{Image length} = |-3f| - |-2f| = f$$

8. Ans (B)

Let x_1 and x_2 be elongations in lower and upper spring

$$\Rightarrow \frac{-x_1 + x}{2} = x_2 \Rightarrow x = 2x_2 + x_1$$

Let T : tension in the string connecting the block.

$$\Rightarrow T = kx_1 \text{ and } T + kx_1 = kx_2$$

$$\Rightarrow 2x_1 = x_2$$

$$\Rightarrow x_1 = x/5 \text{ and } x_2 = 2x/5$$

Now loss in GPE of block = gain in KE + gain in EPEs

$$\Rightarrow mgx = \frac{1}{2}mv^2 + \frac{1}{2}kx_1^2 + \frac{1}{2}kx_2^2$$

$$\Rightarrow \frac{1}{2}mv^2 = mgx - \frac{1}{2}k \left\{ \frac{x^2}{25} + \frac{4x^2}{25} \right\}$$

$$= \frac{10mgx - kx^2}{10}$$

9. Ans (B)

$$f_{\max} = T_{\max}$$

$$4\mu mg = mg + \frac{mv_{\max}^2}{\ell}; v_{\max} = A\sqrt{\frac{g}{\ell}}$$

$$\Rightarrow \mu = \frac{1}{4} \left(1 + \frac{A^2}{\ell^2} \right)$$

12. Ans (B)

$$W = \int 2ydx + \int x^2dy \text{ here } y = x$$

$$W = \int_0^5 2xdx + \int x^2dx$$

$$W = 25 + \frac{125}{3} = \frac{200}{3} = 66.7J$$

14. Ans (B)

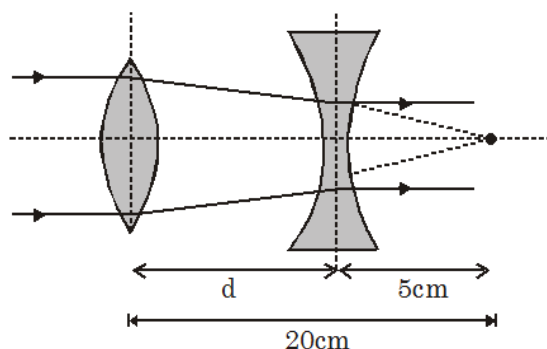
$$\vec{F}_C = - \left[\frac{\partial U}{\partial x} \hat{i} + \frac{\partial U}{\partial y} \hat{j} + \frac{\partial U}{\partial z} \hat{k} \right]$$

$$\text{Put } U = \frac{20yz}{x}$$

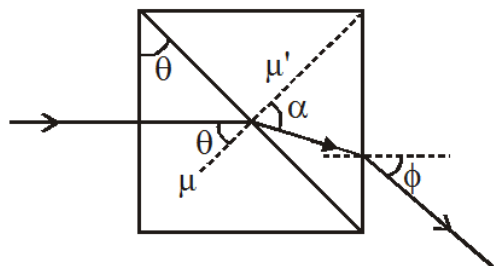
$$\vec{F}_C = \left(\frac{20yz}{x^2} \right) \hat{i} - \left(\frac{20z}{x} \right) \hat{j} - \left(\frac{20y}{x} \right) \hat{k}$$

15. Ans (B)

From figure it is clear that separation between lenses $d = 20 - 5 = 15$ cm



17. Ans (A)



$$\mu \sin \theta = \mu' \sin \alpha$$

$$\Rightarrow \mu \theta = \mu' \alpha \dots (i)$$

$$\mu \sin(\alpha - \theta) = \sin \phi$$

$$\Rightarrow \mu'(\alpha - \theta) = \phi \dots (ii)$$

From (i) and (ii)

$$\mu \theta = \mu' \theta = \phi$$

$$\Rightarrow (\mu - \mu') = \frac{\phi}{\theta}$$

18. Ans (B)

For prism use have,

$$Li + Le = LA + LD \dots (i)$$

if $i = 15^\circ$, then $e = 60^\circ$

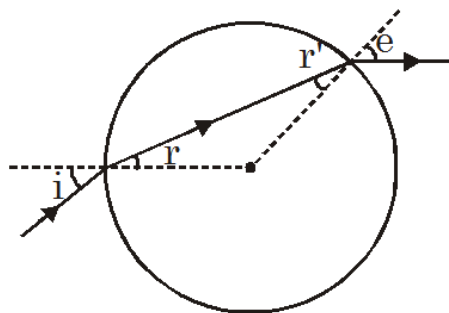
and $D = 30^\circ$

From eq. (i),

$$15^\circ + 60^\circ = A + 30^\circ$$

$$\Rightarrow A = 45^\circ$$

19. Ans (B)



$$\therefore r = r'$$

$$\therefore e = i = 60^\circ$$

$$\text{and } 1 \times \sin 60^\circ = \sqrt{3} \sin r$$

$$\Rightarrow r = 30^\circ$$

$$\therefore \delta = (60^\circ - 30^\circ) + (60^\circ - 30^\circ) = 60^\circ$$

20. Ans (B)

$$\ell = f_0 + f_e, m = \frac{f_0}{f_e}$$

$$\Rightarrow 10 = \frac{20}{f_e} \Rightarrow f_e = 2 \text{ cm}$$

$$\therefore \ell = 20 + 2 = 22 \text{ cm.}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (16)

Charge enclosed

$$q = \int_0^{R/2} Kr 4\pi r^2 dr = \frac{4\pi K}{4} (R/2)^4$$

$$q = \frac{\pi K R^4}{16}$$

$$\text{Using gauss law ; } E 4\pi (R/2)^2 = \frac{\pi K R^4}{16\epsilon_0}$$

$$E = \frac{KR^2}{16\epsilon_0} \text{ So, } N = 16$$

2. Ans (48)

$$v_1 = \frac{20 \times 30}{(20 - 30)} = -60 \text{ cm}$$

$$u \text{ for mirror} = 60 + 10 = 70 \text{ cm}$$

$$v \text{ for mirror} = 70 \text{ cm}$$

$$\therefore u \text{ for lens (again)} = 70 + 10 = 80 \text{ cm}$$

$$\therefore v_2 = \frac{80 \times 30}{(80 - 30)} = 48 \text{ cm}$$

3. Ans (6)

$$v = a\omega$$

$$0.6 = a \times 10$$

$$a = 0.06 \text{ m}$$

4. Ans (5)

$$K \frac{20}{100} = \frac{K10}{100} + mg$$

$$K = 10 \text{ mg}$$

$$T = 2\pi \sqrt{\frac{m}{K_1 + K_2}}$$

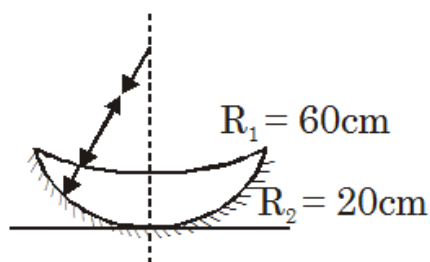
$$T = 2\pi \sqrt{\frac{m}{2K}} = 2\pi \sqrt{\frac{3}{20mg}}$$

$$T = \frac{\pi}{5} \sqrt{\frac{1}{2}} \Rightarrow N = 5$$

5. Ans (4)

Use refraction formula at both surfaces

6. Ans (15)



This silvered concavo-convex lens behaves like a mirror whose focal length can be calculated by the formula

$$\frac{1}{f} = \frac{2}{f_1} + \frac{1}{f_2}$$

f_1 = focal length of concave surface.

f_2 = focal length of concave mirror.

$$\therefore \frac{1}{f} = \frac{2}{-30} + \frac{1}{-10} = \frac{4}{30}$$

$\therefore f = -7.5 \text{ cm}$ Using mirror formula

$$\frac{1}{f} = \frac{1}{v} + \frac{1}{u} \Rightarrow \frac{1}{-7.5} = \frac{1}{-x} + \frac{1}{-x} \Rightarrow x = 15 \text{ cm}$$

8. Ans (40)

$$\frac{D - U}{-U} = -2$$

$$D - U = 2U$$

$$D = 3U$$

$$U = 20$$

$$m = \frac{F}{F - 20}$$

$$-2 = \frac{F}{F - 20}$$

$$-2F + 40 = F$$

$$F = \frac{40}{3} \text{ cm}$$

9. Ans (10)

$$W = W_{OA} + W_{AB}$$

$$= \int_0^1 x^2 dx + \int_0^2 y dy = \frac{1}{3} + \frac{4}{2} = \frac{7}{3} \text{ J}$$

$$\Rightarrow p + q = 10$$

10. Ans (4)

$$T = 2\pi \sqrt{\frac{1}{2mgd}}$$

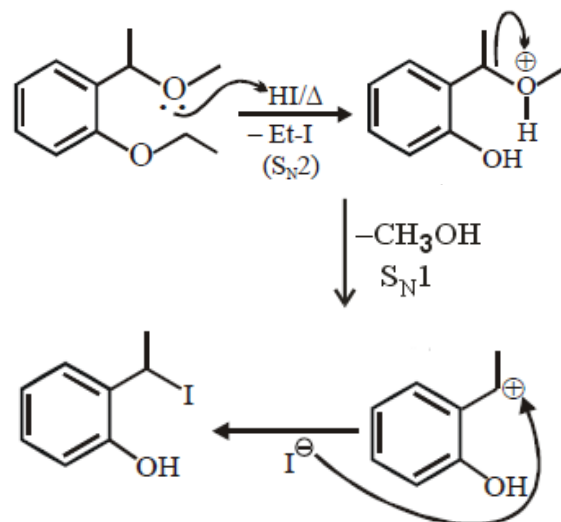
$$= 2\pi \sqrt{\frac{mR^2 + mR^2 + 4mR^2}{2mg \cdot \frac{3}{2}R}}$$

$$= 2\pi \sqrt{\frac{2R}{g}} = 4 \text{ sec}$$

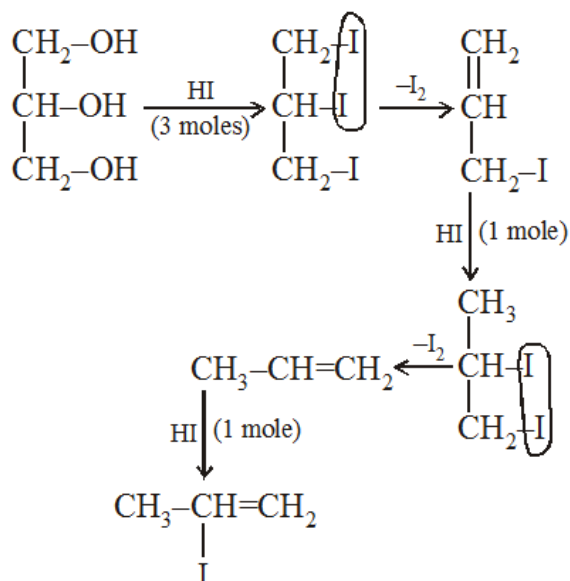
PART-2 : CHEMISTRY

SECTION-I

2. Ans (C)



6. Ans (B)



9. Ans (A)

$$\frac{P_A^0}{P_B^0} = \frac{1}{3}$$

$$\frac{P_A^l}{n_B^l} = \frac{1}{3} \quad y_A = ?$$

$$\therefore X_A = \frac{n_A^l}{n_A^l + n_B^l} = \frac{1}{1+3} = 0.25$$

$$Y_A = \frac{P_A^0 X_A}{P_A^0 X_A + P_B^0 X_B} = \frac{X_A}{X_A + \frac{P_B^0}{P_A^0} X_B}$$

$$\text{or, } Y_A = \frac{0.25}{0.25 + \frac{3}{1} + 0.75} = \frac{0.25}{0.25(1+9)} = \frac{1}{10} = 0.1$$

10. Ans (B)

For hydrolysis of ester

$$r = k[\text{H}^+][\text{ester}]$$

if 0.05 M HCl,

$$k_{\text{HCl}} = k = 0.05 \dots (1)$$

if 0.05 M H₂SO₄

$$k_{\text{H}_2\text{SO}_4} = 1 \times 0.1 \dots (2)$$

$$\frac{(2)}{(1)} \Rightarrow \frac{k_{\text{H}_2\text{SO}_4}}{k_{\text{HCl}}} = \frac{0.1}{0.05} = 2$$

clearly, $k_{\text{H}_2\text{SO}_4} = 2k_{\text{HCl}}$

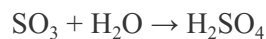
11. Ans (A)

$$\text{Oleum} = 178\text{g}, \left(100 + \frac{900}{89}\right) \%$$

$$\text{for 100 g oleum water required} = \frac{900}{89} \text{ g}$$

$$\therefore 178 \text{ g oleum water required} = \frac{900}{89} \times \frac{178}{100} \text{ g}$$

$$= 18 \text{ g} = 1 \text{ mole}$$

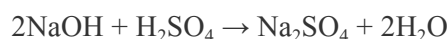


$\therefore$ In 178 g oleum,

$$\text{SO}_3 = 1 \text{ mole} = 80 \text{ g}$$

$$\text{H}_2\text{SO}_4 = 178 - 80 = 98 \text{ g} = 1 \text{ mole}$$

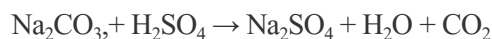
$$\therefore \text{Total H}_2\text{SO}_4 = 1 + 1 = 2 \text{ mole}$$



2mole

$$\therefore \text{NaOH required} = 4 \text{ mole.}$$

12. Ans (B)



$$106 \text{ g} \quad \quad \quad 0.01 \text{ mole}$$

$$= \frac{1.06}{106} \text{ mole} \quad \quad \quad = 0.01 \times 24.6$$

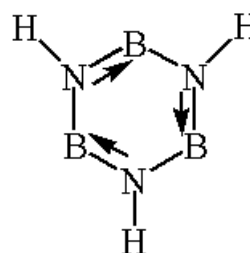
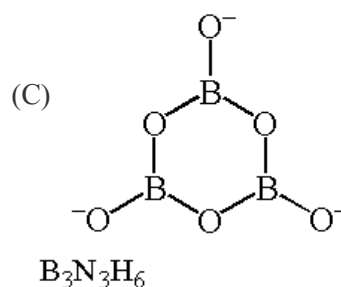
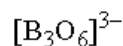
$$= 0.01 \text{ mole} \quad \quad \quad = 0.246$$

13. Ans (C)



14. Ans (C)

(A) CHCl_3 is more acidic than CHF_3 as CCl_3^- is stabilised by back bonding.



Both are planar

PART-2 : CHEMISTRY

SECTION-II

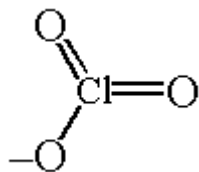
15. Ans (C)

(A) NO_2 odd e^- species

(B) BF_3 hypovalent

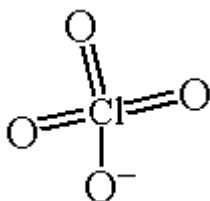
PF_5 hypervalent

(C) ClO_3^-



$$\text{BO} = 1 + \frac{2}{3}$$

$$= \frac{5}{3} = 1.66$$

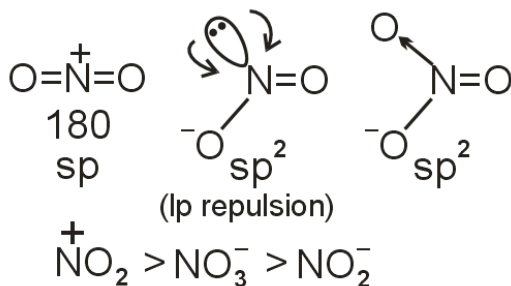


$$\text{BO} = 1 + \frac{3}{4}$$

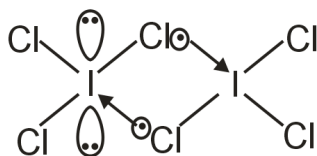
$$= \frac{7}{4} = 1.75$$

16. Ans (A)

(A) $\text{BF}_3 < \text{BCl}_3 < \text{BBR}_3$ [Lewis acidic order]

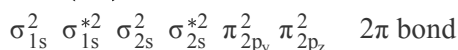


17. Ans (B)



(B) $\text{sp}^3\text{d}^2\text{-sp}^3\text{-sp}^3\text{d}^2$

18. Ans (A)



4. Ans (4)

Number of moles of cation vacancy

$$= \frac{6.02 \times 10^{21}}{6.02 \times 10^{23}} = 10^{-2} \text{ mole}$$

Number of moles of M^{+2} ions

$$= 10^{-2} \text{ mole}$$

Number of moles of missing y^- ions

$= 2 \times 10^{-2}$ mole % decreases in density

$$= \frac{30 \times 10^{-2} - 20 \times 10^{-2}}{25} \times 100\% = 0.4\% = x\%$$

$$\Rightarrow 10x = 4$$

5. Ans (384)

585.5 g solution, 0.5 molal sucrose $\Delta T_f = 4^\circ\text{C}$

1000 g solvent contains 0.5 mole sucrose

$$= 0.5 \times 342 = 171 \text{ g sucrose}$$

$$\therefore \text{mass of solution} = 1000 + 171 = 1171 \text{ g.}$$

$$\therefore 1171 \text{ g solution contains } 0.5 \text{ mole sucrose.}$$

$$\therefore 585.5 \text{ solution contains } \frac{0.5}{1171} \times 585.5 = 0.25$$

mole sucrose.

$$\therefore \text{mass of solvent} = 585.5 - 0.25 \times 342 = 500 \text{ g}$$

Let x g ice is separated out

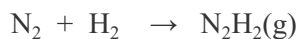
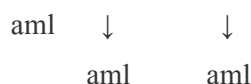
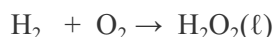
$$\Delta T_f = k_f m$$

$$\text{or, } 4 = \frac{1.86 \times 0.25 \times 1000}{(500 - x)}$$

$$\text{or, } 500 - x = 116.25$$

$$\text{or, } x = 383.75 \text{ g} = 384 \text{ (nearest integer)}$$

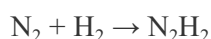
6. Ans (30)



$$\text{O}_2 = (a + 10) \text{ ml}$$

$$\text{contraction in vol} = 60$$

$$\text{or, } 2a + b = 60 \quad \dots (1)$$



cml excess

$$1 \times C = 10$$

$$C \Rightarrow 10 \text{ ml}$$

$$\text{N}_2 = (b + 10) \text{ ml}$$

$$\text{H}_2 = (a + 10) \text{ ml}$$

$$\therefore a + b + b + 10 + a + 10 = 100$$

$$\text{or, } 2a + 2b = 80$$

$$\text{or, } a + b = 40 \text{ ml} \quad \dots (2)$$

$$(1) + (2) \Rightarrow a = 20 \text{ ml} \therefore b = 20 \text{ ml}$$

$$\therefore \text{N}_2 = 30 \text{ ml}, \text{H}_2 = 40 \text{ ml}, \text{O}_2 = 30 \text{ ml}$$

$$\therefore \text{mole \% of N}_2 = \frac{30}{100} \times 100 = 30\%$$

7. Ans (8)

$$\text{dipole moment} = 1.6 \text{ D} = \mu_0$$

$$\text{Bond length} = 1.625 \text{ \AA}$$

$$\% \text{ionic} = \frac{\mu_{\text{obs}}}{\mu_{\text{theo}}} \times 100 = \frac{1.6}{q \times d} \times 100$$

$$= \frac{1.6 \text{ D}}{(1.6 \times 10^{-19})(1.625 \times 10^{-10})} \times 100$$

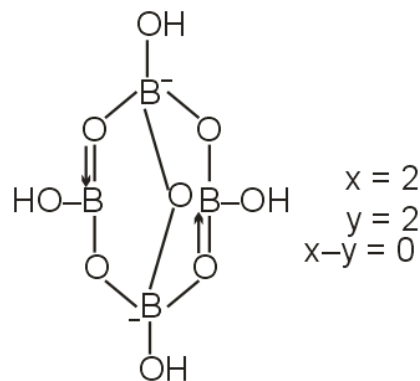
$$= \frac{1.6 \times 3.3 \times 10^{-30}}{(1.6 \times 10^{-19}) \times (1.625 \times 10^{-10})} \times 100$$

$$x = \frac{3.3}{1.625} \times 10 = \frac{33}{1.625} = 20.3$$

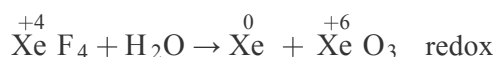
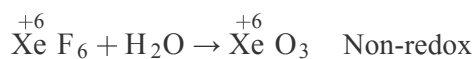
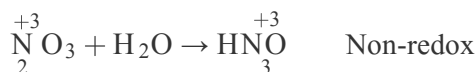
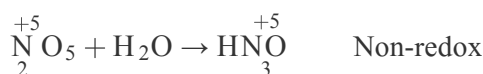
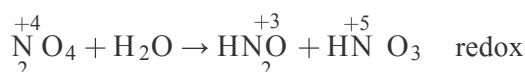
$$x \times 0.39$$

$$20.3 \times 0.39 \approx 8$$

8. Ans (0)



9. Ans (3)



PART-3 : MATHEMATICS

SECTION-I

1. Ans (A)

L.H.S is an even integer & R.H.S. is an odd integer.

2. Ans (D)

$$3\sin x + 4\cos x = \frac{10y + 2}{1 - y}$$

$$\Rightarrow -5 \leq \frac{10y + 2}{1 - y} \leq 5$$

$$\Rightarrow y \in \left[-\frac{7}{5}, \frac{1}{5} \right]$$

3. Ans (C)

$$\text{RHL} = \lim_{h \rightarrow 0} \frac{\sin^{-1}(\sin(0 + h))}{\cos^{-1}(\cos(0 + h))} = \lim_{h \rightarrow 0} \frac{h}{h} = 1$$

$$\text{LHL} = \lim_{h \rightarrow 0} \frac{\sin^{-1}(\sin(0 - h))}{\cos^{-1}(\cos(0 - h))} = \lim_{h \rightarrow 0} \frac{-h}{h} = -1$$

$\therefore \text{LHL} \neq \text{RHL} \therefore \text{Limit doesn't exist.}$

4. Ans (B)

Clearly $f(x)$ is not differentiable at $x = \frac{3}{2}$ and 2

5. Ans (A)

$$\begin{aligned} \text{LHL} &= \lim_{x \rightarrow 0^-} (a \cot^2 x \ln \cos x) \\ &= \lim_{x \rightarrow 0^-} \frac{a \ln (1 + (\cos x - 1))}{\tan^2 x} \\ &= \lim_{x \rightarrow 0^-} \frac{a \ln (1 + (\cos x - 1))}{(\cos x - 1)} \cdot \frac{\tan^2 x}{x^2} \\ &= \lim_{x \rightarrow 0^-} a \left(\frac{\cos x - 1}{x^2} \right) = -\frac{a}{2} \end{aligned}$$

$$f(0) = 1$$

RHL

$$\begin{aligned} &= \lim_{x \rightarrow 0^+} (\sec x)^{b \cot x} = \lim_{x \rightarrow 0^+} e^{b \cot x (\sec x - 1)} \\ &= e^{\lim_{x \rightarrow 0^+} \left\{ b \left(\frac{1 - \cos x}{\sin x} \right) \right\}} = e^0 = 1 \quad (\text{where, } b \in \mathbb{R}) \end{aligned}$$

$\therefore a = -2$ and $b \in \mathbb{R}$ Ans.

6. Ans (C)

$$\text{Let } L = \lim_{x \rightarrow 0} \left(\frac{p^x + q^x + r^x}{3} \right)^{\frac{1}{x}} \rightarrow (1^\infty \text{ form})$$

$$\begin{aligned} &= e^{\lim_{x \rightarrow 0} \frac{1}{x} \left(\frac{p^x + q^x + r^x}{3} - 1 \right)} \\ &= e^{\lim_{x \rightarrow 0} \frac{1}{3} \left[\left(\frac{p^x - 1}{x} \right) + \left(\frac{q^x - 1}{x} \right) + \left(\frac{r^x - 1}{x} \right) \right]} \end{aligned}$$

$$= e^{\frac{1}{3} [\ell np + \ell nq + \ell nr]}$$

$$= e^{\frac{1}{3} \ell n(pqr)} = (pqr)^{1/3}$$

Now apply A.M. $\geq$ G.M.

$$\frac{p + q + r}{3} \geq (pqr)^{1/3} \Rightarrow \frac{21}{3} \geq L \Rightarrow L \leq 7$$

$\therefore L$ cannot be 8

7. Ans (D)

$$\begin{aligned} \frac{d^2 x}{dy^2} &= - \frac{\left(\frac{d^2 y}{dx^2} \right)}{\left(\frac{dy}{dx} \right)^3} \\ &\Rightarrow \frac{\left(\frac{d^2 y}{dx^2} \right)}{\left(\frac{d^2 x}{dy^2} \right)} = - \left(\frac{dy}{dx} \right)^3 = - \left(\frac{\cos t}{-\sin t} \right)^3 \\ &= \cot^3 t \text{ Ans.} \end{aligned}$$

8. Ans (B)

$$y_1 = y' = \cos x - \sin x$$

$$y_2 = y'' = -\sin x - \cos x$$

$$y_3 = y''' = -\cos x + \sin x$$

$$y_4 = y'''' = \sin x + \cos x = y$$

$$y_4 = y$$

Derivative of y repeats itself after every 4 derivatives.

$$\therefore f(x) = \frac{d^{2020} y}{dx^{2020}} = y_{2020} = y$$

$$\begin{aligned} \therefore y \left(\frac{\pi}{4} \right) f \left(\frac{5\pi}{4} \right) &= y \left(\frac{\pi}{4} \right) \cdot y \left(\frac{5\pi}{4} \right) \\ &= \left(\sin \frac{\pi}{4} + \cos \frac{\pi}{4} \right) \cdot \left(\sin \frac{5\pi}{4} + \cos \frac{5\pi}{4} \right) \\ &= \left(\frac{1}{\sqrt{2}} + \frac{1}{\sqrt{2}} \right) \left(\frac{-1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right) \\ &= -2 \end{aligned}$$

9. Ans (B)

$$x^2 + y^2 = \left(t + \frac{1}{t} \right)^2 - 2$$

$$\Rightarrow x^2 + y^2 = (x + y)^2 - 2$$

$$\Rightarrow xy = 1$$

$$\Rightarrow y = \frac{1}{x}$$

$$\Rightarrow \frac{dy}{dx} = -\frac{1}{x^2}$$

$$\therefore x^2 \frac{dy}{dx} = -1 \text{ Ans.}$$

10. Ans (A)

$$y' = e^{\tan^{-1} x} \left(\frac{1}{1 + x^2} \right) + e^{-\cot^{-1} x} \left(\frac{1}{1 + x^2} \right)$$

$$\Rightarrow y' = \left(e^{\tan^{-1} x} + e^{-\cot^{-1} x} \right) \left(\frac{1}{1 + x^2} \right)$$

$$\Rightarrow y' = \frac{y}{1 + x^2}$$

$$\Rightarrow (1 + x^2)y' = y$$

$$\Rightarrow (2x)y' + (1 + x^2)y'' = y'$$

$$\Rightarrow (1 + x^2)y'' + y'(2x - 1) = 0$$

11. Ans (A)

$$\text{Let } y = \sin^2 \left(\cot^{-1} \sqrt{\frac{1+x}{1-x}} \right).$$

$$\text{Put } x = \cos 2\theta \quad \theta \in \left(0, \frac{\pi}{2} \right]$$

$$\therefore y = \sin^2 \cot^{-1} \left(\sqrt{\frac{1+\cos 2\theta}{1-\cos 2\theta}} \right) = \sin^2 \cot^{-1} (\cot \theta)$$

$$\therefore y = \sin^2 \theta = \frac{1 - \cos 2\theta}{2} = \frac{1 - x}{2} = \frac{1}{2} - \frac{x}{2}$$

$$\therefore \frac{dy}{dx} = -\frac{1}{2}.$$

12. Ans (D)

$$g(x) = f^{-1}(x)$$

$$\Rightarrow g(f(x)) = x$$

$$\Rightarrow g'(f(x)) \cdot f'(x) = 1$$

$$\Rightarrow g'(f(x)) = \frac{1}{f'(x)}$$

$$\Rightarrow g''(f(x)) \cdot f'(x) = -\frac{1}{(f'(x))^2} \cdot f''(x)$$

$$\Rightarrow g''(f(x)) = \frac{-f''(x)}{(f'(x))^3}$$

$$g''(x^3 + x) = \frac{-(6x)}{(3x^2 + 1)^3}$$

$$\text{Put } x = -1 \Rightarrow g''(-2) = -\frac{6(-1)}{(3+1)^3} = \frac{3}{32}$$

13. Ans (D)

$$f(x) = \cot^{-1}(\sqrt{x^2-1}) + \sec^{-1}\left(\frac{x}{\sqrt{x^2-1}}\right) + \cos \sec^{-1}x$$

If $x < -1$,

$$f(x) = -\sin^{-1}\left(\frac{1}{x}\right) + \left(\pi + \sin^{-1}\left(\frac{1}{x}\right)\right) + \sin^{-1}\left(\frac{1}{x}\right)$$

$$f(x) = \pi + \sin^{-1}\left(\frac{1}{x}\right)$$

$$\therefore f'(x) = 0 + \frac{1}{\sqrt{1-\frac{1}{x^2}}} \left(-\frac{1}{x^2}\right)$$

$$\therefore f'(-2) = \frac{1}{\sqrt{1-\frac{1}{4}}} \left(-\frac{1}{4}\right) = -\frac{1}{2\sqrt{3}}$$

14. Ans (C)

$$\lim_{x \rightarrow \frac{1}{3}} (f(x))^{\frac{1}{3x-1}} \quad (\because 1^\infty)$$

$$\Rightarrow f\left(\frac{1}{3}\right) = 1$$

$$\Rightarrow \frac{1}{27} + \frac{a}{9} + \frac{a}{3} + 1 = 1$$

$$a = -\frac{1}{12}$$

Now sum of roots of equation $f(x) = 0$ is $-a$

15. Ans (B)

Differentiate given rule w.r.t. y treating x as constant

$$f'(x+y) = f'(y) + 3x^2 + 6xy$$

at $y = 0$

$$f'(x) = f'(0) + 3x^2 + 0 = 1 + 3x^2 > 0$$

$$\therefore f(x) = x^3 + x + c$$

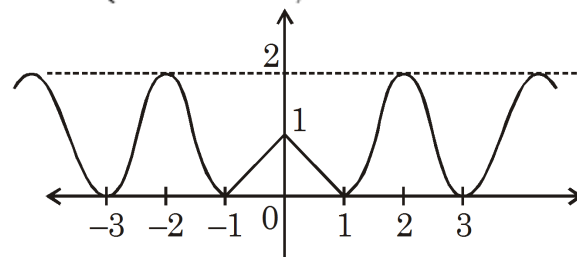
$$\Rightarrow f(0) = 0 + 0 + c$$

$$\Rightarrow 2 = c \quad (\text{at } x = 0, y = 0 \Rightarrow f(0) = 2)$$

$$\therefore f(x) = x^3 + x + 2 \text{ which is one-one}$$

16. Ans (C)

$$f(x) = \begin{cases} 1 - |x| & ; x^2 < 1 \\ 0 & ; x^2 = 1 \\ 1 + \cos \pi x & ; x^2 > 1 \end{cases}$$



Non-differentiable at $x = -1, 0, 1$

17. Ans (C)

$$f_2(Jo(f_1(x))) = f_3(x)$$

$$\Rightarrow 1 - J(f_1(x)) = f_3(x)$$

$$\Rightarrow 1 - J\left(\frac{1}{x}\right) = \frac{1}{1-x}$$

$$\Rightarrow J\left(\frac{1}{x}\right) = 1 - \frac{1}{1-x} = -\frac{x}{1-x}$$

$$\Rightarrow J(x) = \frac{-\frac{1}{x}}{1 - \frac{1}{x}} = \frac{1}{1-x} \Rightarrow J(x) = f_3(x)$$

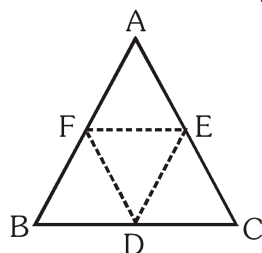
18. Ans (C)

$$\begin{aligned} & \frac{\sum \frac{\Delta}{s-a}}{\sqrt{\sum \frac{\Delta^2}{(s-a)(s-b)}}} \\ &= \frac{\sum \frac{1}{s-a}}{\sqrt{\sum \frac{1}{(s-a)(s-b)}}} \\ &= \frac{\sum \frac{1}{s-a}}{\sqrt{\sum \frac{s(s-c)}{\Delta^2}}} \\ &= \sum \frac{1}{(s-a)} \times \frac{\Delta}{s} = \sum \tan \frac{A}{2} \\ &(\because r = (s-a) \tan \frac{A}{2} = (s-b) \tan \frac{B}{2} = (s-c) \tan \frac{C}{2}) \end{aligned}$$

19. Ans (B)

We know

$$AD^2 + BE^2 + CF^2 = \frac{3}{4}(a^2 + b^2 + c^2)$$



$$\text{also } FE = \frac{a}{2}, DE = \frac{c}{2} \text{ \& } DF = \frac{b}{2}$$

$$\Rightarrow \frac{AD^2 + BE^2 + CF^2}{EF^2 + DE^2 + DF^2} = \frac{3}{1}$$

20. Ans (A)

$$\frac{2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x}}{3} > (2^{\cos^{-1}x + \sin^{-1}x + \tan^{-1}x})^{1/3}$$

$$2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3 \cdot \left(2^{\frac{\pi}{2} + \tan^{-1}x}\right)^{1/3}$$

$$\Rightarrow 2^{\sin^{-1}x} + 2^{\cos^{-1}x} + 2^{\tan^{-1}x} > 3$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (4)

$$f' \left(\frac{\pi}{2} \right) = \begin{vmatrix} -1 & 1 & 0 \\ 0 & 2 & -2 \\ 3 & -1 & 0 \end{vmatrix} + 0 + \begin{vmatrix} 0 & 1 & -1 \\ -1 & 0 & 0 \\ 0 & -1 & 9 \end{vmatrix}$$

2. Ans (5)

Let $f(x) = y$

$$\therefore y = \frac{\sin x}{1 + \frac{\cos x}{1 + \frac{\sin x}{1 + \frac{\cos x}{1 + \dots}}}}$$

$$\Rightarrow y = \frac{\sin x}{1 + \left(\frac{\cos x}{1+y} \right)}$$

$$\Rightarrow y = \frac{(1+y) \sin x}{1+y+\cos x}$$

$$\Rightarrow y + y^2 + y \cos x = \sin x + y \sin x \quad \dots (1)$$

Put $x = 0$, we get $y + y^2 + y = 0 + 0$

$$\Rightarrow y^2 + 2y = 0 \Rightarrow y = 0, -2$$

✓ ✗

$\therefore y = 0$ at $x = 0$

Now differentiate eq. (1) w.r.t. x

$$\frac{dy}{dx} + 2y \frac{dy}{dx} + \cos x \cdot \frac{dy}{dx} - (\sin x) y$$

$$= \cos x + \frac{dy}{dx} \sin x + y \cos x$$

Put $x = 0, y = 0$

$$\Rightarrow \frac{dy}{dx} + 0 + \cos 0 \cdot \frac{dy}{dx} - 0 = 1 + 0 + 0$$

$$\Rightarrow 2 \frac{dy}{dx} = 1 \Rightarrow \frac{dy}{dx} = \frac{1}{2}$$

$$\therefore 10f'(0) = 5$$

3. Ans (9)

$$f'(x) \cdot f(-x) - f(x) f'(-x) = 0$$

$$\Rightarrow \frac{d}{dx} (f(x) \cdot f(-x)) = 0$$

$$\Rightarrow f(x) \cdot f(-x) = k$$

$$\text{Put } x = 0 \Rightarrow f(0) \cdot f(0) = k \Rightarrow k = 9$$

$$\therefore f(x) \cdot f(-x) = 9$$

$$\therefore f(3) \cdot f(-3) = 9 \text{ Ans.}$$

5. Ans (2)

$$\text{fog}(x) = \text{gof}(x)$$

$$\Rightarrow \frac{10^{x+1} - 1}{10^{x+1} + 1} = \frac{10^x - 1}{10^x + 1} + 1 = \frac{2 \cdot 10^x}{10^x + 1}$$

use C & D

$$\Rightarrow 10^{x+1} = \frac{3 \cdot 10^x + 1}{1 - 10^x} \quad \text{Let } 10^x = t$$

$$\Rightarrow 10t^2 - 7t + 1 = 0 \Rightarrow t = \frac{1}{2} \text{ or } \frac{1}{5}$$

6. **Ans (2)**

$$f(x) = \{x\} + \{2x\} + \{3x\}$$

$\{x\}$ is not differentiable at $x \in \mathbb{N}$

$$\Rightarrow x = \{1, 2, 3, \dots\}$$

$\{2x\}$ is not differentiable at $2x \in \mathbb{N}$

$$\Rightarrow x = \left\{ \frac{1}{2}, 1, \frac{2}{2}, 2, \frac{5}{2}, \dots \right\}$$

$\{3x\}$ is not differentiable at $3x \in \mathbb{N}$

$$\Rightarrow x = \left\{ \frac{1}{3}, \frac{2}{3}, 1, \frac{4}{3}, \frac{5}{3}, 2, \dots \right\}$$

$\therefore f(x)$ is N.D. at

$$x \in \left\{ \frac{1}{3}, \frac{1}{2}, \frac{2}{3}, 1, \frac{4}{3}, \frac{3}{2}, \frac{5}{3}, 2, \dots \right\}$$

$\therefore f(x)$ is N.D. at 7 points in $[0, 2)$

$\therefore n = 2$ Ans.

7. **Ans (1)**

as we know that

$$\frac{a}{\sin A} = 2R \text{ and } R = 2R_p$$

$$\frac{2}{\sin 30^\circ} = 2R$$

$$R = 2 \Rightarrow R_p = 1$$

8. **Ans (2)**

$$\because 2 \leq a < 3 \Rightarrow [a] = 2$$

$$\therefore \cos^{-1}\cos 2 + \operatorname{cosec}^{-1}\operatorname{cosec} 2 + \cot^{-1}\cot 2$$

$$= 2 + \pi - 2 + 2 = \pi + 2$$

9. **Ans (3)**

$$\sin^2 x + \cot^2 x = 3$$

$$\text{Let } s = \sin^2 x$$

$$\Rightarrow s + \frac{1-s}{s} = 3 \Rightarrow s^2 + 1 - s = 3s$$

$$\Rightarrow s^2 - 4s + 1 = 0 \Rightarrow s = \frac{4 \pm 2\sqrt{3}}{2}$$

$$\Rightarrow \sin^2 x = 2 - \sqrt{3} \Rightarrow \operatorname{cosec}^2 x = 2 + \sqrt{3}$$

$$\Rightarrow [\operatorname{cosec}^2 x] = 3$$

10. **Ans (3)**

$$\sin^2 \theta - 2\sin \theta + \sin^2 \alpha - 2\sin \alpha = 0$$

$$(1 - \sin \theta)^2 + (1 - \sin \alpha)^2 = 2$$

Possible only when $\sin \theta = \sin \alpha = 0$

$$\theta + \alpha = 0 + 0 \text{ or } \theta + \alpha = 0 + \pi$$

$$\text{or } \theta + \alpha = \pi + \pi$$

number of possible values are three.